

WORKSHEET A.5.M1

Introduction

What we will learn today:

- Explain what a network is and identify its categories.
- Identify the equipment needed to connect a device to a network.
- Distinguish between connection media: cables, fibre optics and wireless connections.
- Recognise basic network devices (switch, router and access point) and describe their roles.
- State advantages and disadvantages of wireless networks.
- Name common network services, such as file sharing, shared printing, email and the World Wide Web.
- Describe the client-server communication model and identify the client and server in examples.

Consolidation / Assessment Activities

1. Decide whether each statement is true or false:

No.	Statement	True	False
(a)	A computer network is a group of two or more computers and other devices connected so that they can exchange data.	☐	☐
(b)	Our school network is a wide area network.	☐	☐
(c)	Our home network is a local area network.	☐	☐
(d)	To create a computer network, we need networking equipment such as cables, network cards, switches and routers.	☐	☐
(e)	To connect the computers in a computer lab, we need at least one switch.	☐	☐
(f)	A router is required to connect a bank branch network to the head-office network.	☐	☐
(g)	For every computer on a network to print, each computer must have its own printer.	☐	☐
(h)	A local network cannot provide access to files, folders or programs stored on other computers.	☐	☐
(i)	When you read email, your computer acts as the server and the email provider's computer acts as the client.	☐	☐

2. Name devices other than computers - for example, an alarm system - that can be connected to a network.

3. Identify the network devices and connection media shown below. Write the number of the corresponding image next to each name:

- (a) Fibre optics _____
- (b) Switch _____
- (c) Network cables (Cat 5, Cat 6) _____
- (d) Modem _____
- (e) Router _____
- (f) Ethernet card _____

(1)



(2)



(3)



(4)



(5)



(6)



4. For each case, state the most suitable connection medium: fibre optics, network cable or wireless connection.

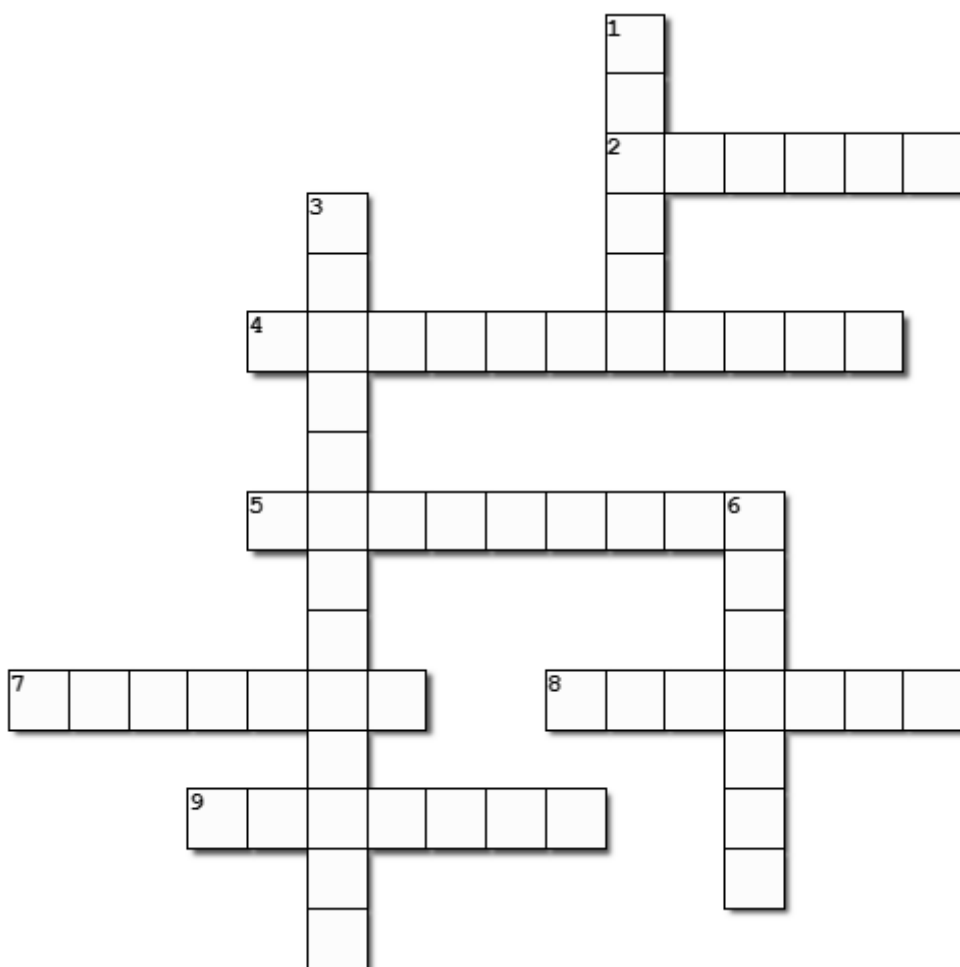
- (a) Connecting a home network to the provider _____
- (b) Connecting computers in a computer lab to the switch in the room _____
- (c) Connecting a tablet to the home network _____
- (d) Connecting a desktop computer to the office network _____
- (e) Connecting the computer lab to the router _____
- (f) An undersea connection between countries _____

5. State two advantages and two disadvantages of wireless networks.

6. Name two Internet services.

Additional Activities

7. Complete the crossword below.



ACROSS

2. The _____ of fibre-optic cables is greater than that of ordinary network cables (Cat 5, Cat 6).
4. To connect our home network to the Internet, a _____ is required.
5. A _____ connects two or more computers in a network so that they can exchange information.
7. Cat 5 and Cat 6 are special _____ that connect a computer network card to another computer or a switch.
8. One network category is a _____ area network.
9. A computer must have a _____ card to connect to a network.

DOWN

1. A _____ is a group of two or more connected computers.
3. One major advantage of networks is sharing _____ devices.
6. One major advantage of networks is the sharing of _____.
8. Name the additional devices required to create a network using five computers and one printer that you already have. Justify your answer.

Activities for Home

9. What is a network?

10. Ms Maria manages an architectural office. Three architects and one interior designer work there. They use computers, printers and a plotter every day, and often need to exchange files.

(a) Explain why installing a network could improve the quality and speed of the office's work.

(b) Explain which additional devices are required to create the network.
